

Maximising Edge Density in Graphs Subject to Connectivity Constraints

Erdős Problem 915, from Proof to Search

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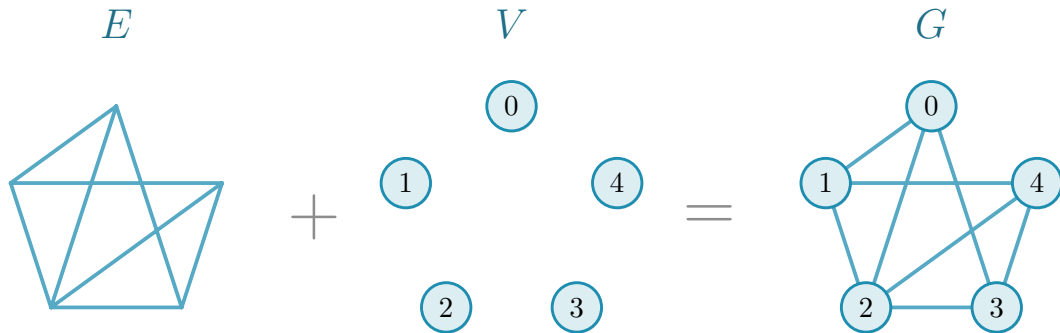
Maximising edge density in graphs
under connectivity constraints



translation

networks with **many links** and **low connectivity**

Graph

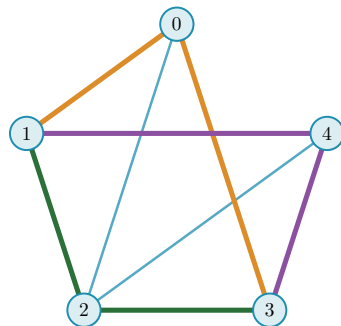


Erdős Problem 915

$$\ell_m(n) = \max_{|V(G)|=n} \left\{ |E(G)| : \lambda_G^{\max} \leq m-1 \right\}$$

$$\lambda_G^{\max} = \max_{u \neq v} \lambda_G(u, v)$$

$\lambda_G(u, v)$ = the largest number of u - v routes that share **no edge**: one colour each

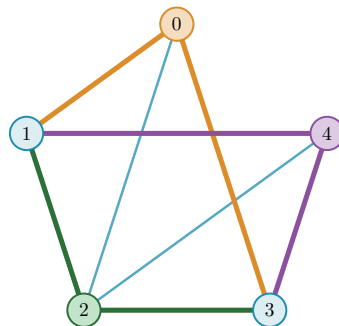


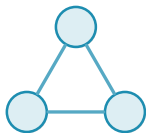
The same question, vertex by vertex

$$k_m(n) = \max_{|V(G)|=n} \{ |E(G)| : \kappa_G^{\max} \leq m-1 \}$$

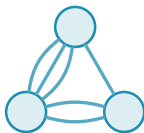
$$\kappa_G^{\max} = \max_{u \neq v} \kappa_G(u, v)$$

$\kappa_G(u, v)$ = the largest number of u - v routes that share **no interior vertex**

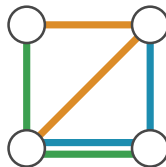




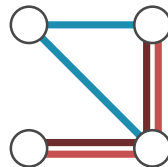
simple graph



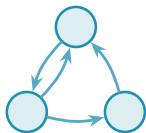
multigraph



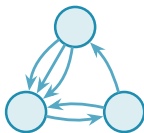
hypergraph



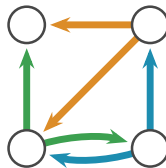
multihypergraph



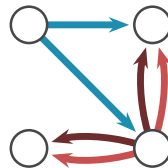
simple digraph



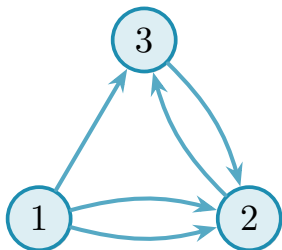
directed multigraph



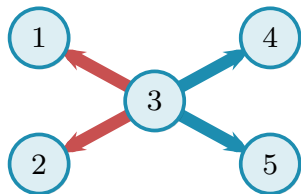
directed hypergraph



directed multihypergraph



$$\begin{pmatrix} \mathbf{0} & 2 & 1 \\ 0 & \mathbf{0} & 1 \\ 0 & 1 & \mathbf{0} \end{pmatrix}$$



list of hyperedges:

$$e_1 : \{3 \mid 1, 2\}$$

$$e_2 : \{3 \mid 4, 5\}$$

$$\begin{pmatrix} 0 & 2 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

INPUT

||

directed multigraph

max flow
over every pair

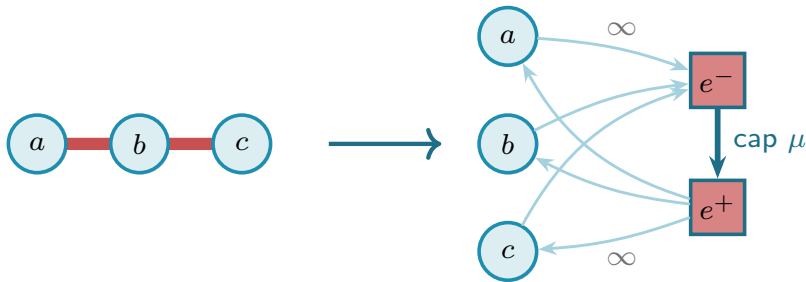
$$\lambda_G^{\max} = 3$$

OUTPUT

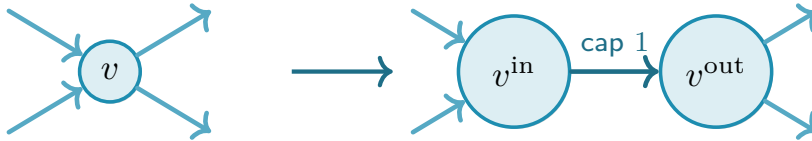
undirected $\rightarrow$ directed



hypergraph $\rightarrow$ non-hypergraph



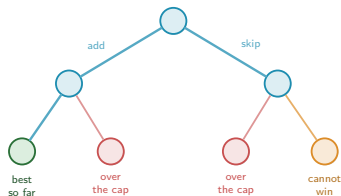
$$\kappa \rightarrow \lambda$$



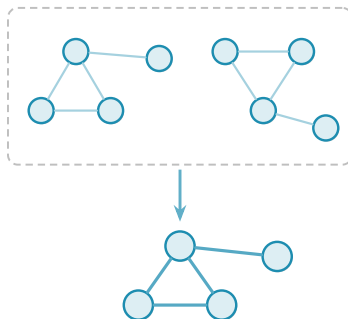
every intermediate vertex is split this way
the flow itself runs from s^{out} to t^{in}

Efficiency: three ways to make the sweep finish

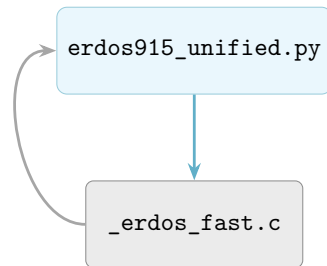
pruning



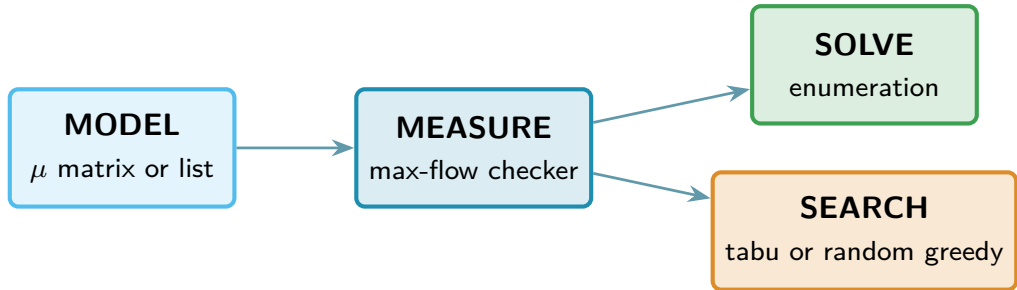
generating search



C: faster max flow

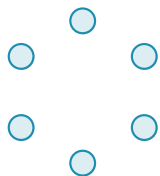


Solving versus searching

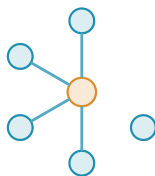


The search: one number per graph

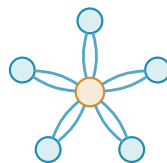
$$\Phi(G) = \underbrace{-|E(G)|}_{\substack{\text{reward} \\ \text{more edges, lower energy}}} + \underbrace{\Lambda \cdot \max(0, c^{\max}(G) - (m - 1))}_{\substack{\text{penalty} \\ \text{discourages excess connectivity}}}$$



$$\Phi = 0$$



$$\Phi = -4$$



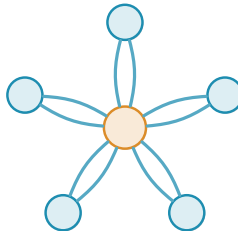
$$\Phi = -10 \text{ at } m = 3$$

Extremal families the search rediscovers



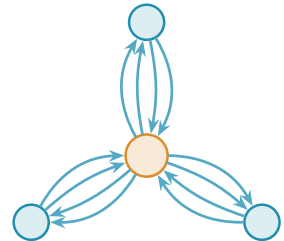
tree

$$n = 6, \ell_2(n) = 5$$



multigraph star

$$n = 6, m = 3, L_3(n) = 10$$

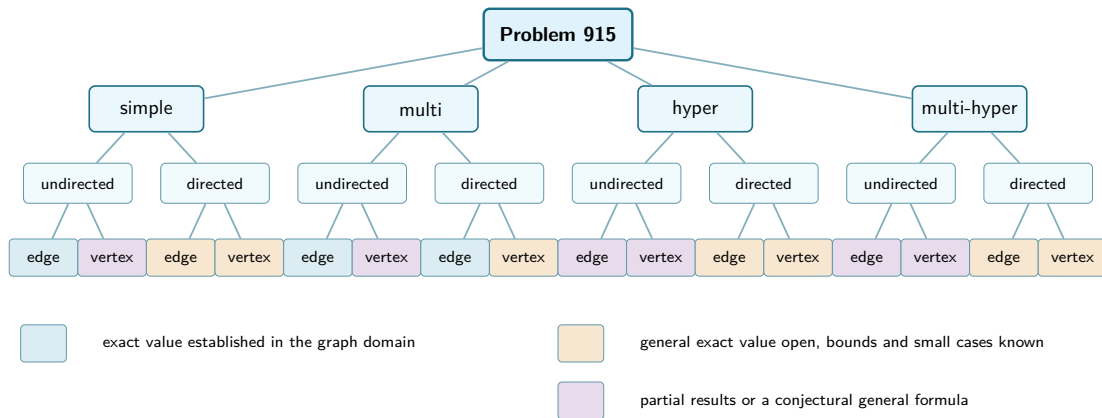


double star

$$n = 4, m = 3, L_3^{\text{dir}}(n) = 12$$

examples of the families, not the saved search witnesses

Status of the sixteen variants



Thank you

`github.com/louisivandenbruwaene/thesis`

Backup: the sixteen variants

HOW TO READ THE NEXT SIXTEEN FRAMES

Each frame fixes one small size and gives the two bounds there as numbers, rather than a rule for general n and m .

UPPER the number, and in grey beside it why nothing can beat it

LOWER the number, and in grey what attains it

EXTREMAL the drawing is that witness, at that size, so every vertex and every edge on it can be counted

Simple graph · edge

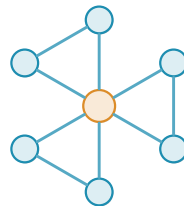
$$\ell_m(n)$$

UPPER 9 **PROVED** $n - 1$ cuts, each of capacity at most $m - 1$

LOWER 9 **PROVED** a star with a matching on its leaves

- the Gomory–Hu tree summarises every pairwise connectivity in $n - 1$ cuts, and charging each edge of the graph to one of them counts every edge twice
- that second charge is what simplicity buys, and it is the whole reason the count is halved

SCOPE Exact at every size with $n \geq m$. Below that not even the complete graph carries m routes, so every edge is allowed.



$n = 7, m = 3$ **EXTREMAL**

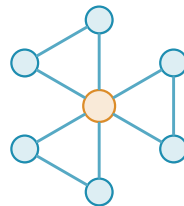
Simple graph · vertex

$$k_m(n)$$

UPPER 9 **PROVED** proved equal to the edge value while $m \leq 4$

LOWER 9 **PROVED** the same graph: every leaf has degree two

- the edge bound does not simply transfer, because a vertex-feasible graph is the weaker restriction and there are more of them
- in the drawing no pair can beat two routes on degree grounds alone, which is why the same graph settles both problems here



$n = 7, m = 3$ **EXTREMAL**

SCOPE Exact for $m \leq 4$ at every n . At $m = 5$ the two agree up to $n = 13$ and first differ at $n = 14$.

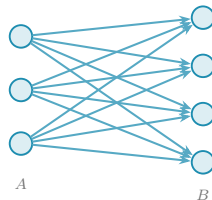
Simple digraph · arc

$$\ell_m^{\text{dir}}(n)$$

UPPER 12 **PROVED** at $m = 2$ a feasible digraph is its own skeleton

LOWER 12 **PROVED** every arc from three tails to four heads

- a reachability skeleton is a subdigraph preserving who reaches whom, and at $m = 2$ nothing can be spared: drop an arc and the route replacing it would be a second disjoint one, breaching the cap
- a skeleton holds at most $\max(2(n-1), \lfloor n^2/4 \rfloor)$ arcs, and at seven vertices those two branches tie, so a bidirected hub does just as well



$n = 7, m = 2$ **EXTREMAL**

SCOPE Exact at every n for $m = 2$. For $m \geq 3$ only the growth rate is settled, and the exact values stay open.

Simple digraph · vertex

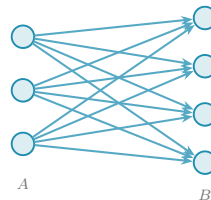
$$k_m^{\text{dir}}(n)$$

UPPER 12 **PROVED** the skeleton argument is already a vertex argument

LOWER 12 **PROVED** the same digraph

- the two routes that argument produces are internally vertex-disjoint, not merely arc-disjoint, so it proves the vertex bound first and the arc bound follows
- that direction matters: Whitney's inequality alone would not justify carrying a bound the other way

SCOPE Exact at every n for $m = 2$. Whether arc and vertex agree for $m \geq 3$ is open.



$n = 7, m = 2$ **EXTREMAL**

Multigraph · edge

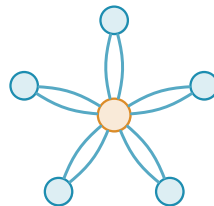
$$L_m(n)$$

UPPER 10 **PROVED** the same $n - 1$ cuts, with no second charge to halve

LOWER 10 **PROVED** a spanning tree at multiplicity $m - 1$

- the same charge to the Gomory–Hu tree runs, but parallel copies mean nothing forces an edge to be counted twice, so the bound is a clean product rather than a halved one
- the missing factor of two is exactly what simplicity was buying next door, and any tree shape does as well as the star

SCOPE Exact at every n and every m .



$n = 6, m = 3$ **EXTREMAL**

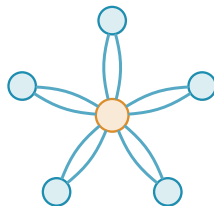
$$K_m(n)$$

UPPER 10 **PROVED** routes split into direct copies plus detours

LOWER 10 **PROVED** a doubled tree: a tree edge has no detour

- between two vertices the count is exactly the multiplicity of their own edge plus the number of disjoint detours around it, so feasibility caps those two together at $m - 1$
- on a tree edge there is no detour at all, which is what lets the multiplicity run all the way up to $m - 1$

SCOPE Exact at $m = 2$ and $m = 3$ for every n . Beyond that only single sizes are settled, such as 14 at $n = 4$, $m = 5$ by exhaustion.



$n = 6, m = 3$ **EXTREMAL**

Directed multigraph · arc

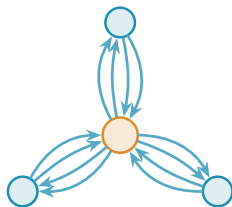
$$L_m^{\text{dir}}(n)$$

UPPER 12 **PROVED** peel off $m - 1$ skeletons, one at a time

LOWER 12 **PROVED** a bidirected tree at multiplicity $m - 1$

- removing one copy of every skeleton arc costs each reachable pair one route, so $m - 1$ peels account for every arc in the digraph
- it is the $m = 2$ skeleton again, now used $m - 1$ times over, which is why the answer is that value multiplied by $m - 1$

SCOPE Exact at every n and every m , and a single threshold-free optimisation settles all of them at once.



$n = 4, m = 3$ **EXTREMAL**

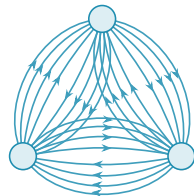
Directed multigraph · vertex

$$K_m^{\text{dir}}(n)$$

UPPER 24 **PROVED** direct copies plus detours, once more

LOWER 24 **PROVED** four copies on each of the six directions

- with only one other vertex available, an ordered pair has exactly one detour, so its multiplicity stops at four rather than at the cap of five, and 6×4 gives the answer
- the arc extremiser does not reach it: twenty arcs against twenty-four, since vertex feasibility is the weaker restriction



$$n = 3, m = 6 \quad \text{EXTREMAL}$$

SCOPE Enumerating every directed multigraph on three vertices leaves this one isomorphism class. Larger sizes stay open.

Simple hypergraph · edge

$$\ell_m^{(r)}(n)$$

UPPER 4 **PROVED** induction on a hyperedge cut, with no tree at all

LOWER 4 **PROVED** a hub and four disjoint pairs of leaves

- a hyperedge spans r vertices but still serves only one route, and cutting one off divides what remains by $r - 1$
- the reduction to graphs points the wrong way: replacing a hyperedge by a star on its vertices would let two routes take two of those edges, which Berge routes forbid

SCOPE Exact whenever there are enough distinct hyperedges to go round, which holds here. It can fail at the smallest sizes, such as $n = r = m = 3$.



$$n = 9, r = 3, m = 2$$

EXTREMAL

Simple hypergraph · vertex

$$k_m^{(r)}(n)$$

UPPER 4 **PROVED** at $m = 2$ the incidence graph must be a forest

LOWER 4 **PROVED** the same hypergraph: its incidence graph is a tree

- a cycle in the incidence graph would be two disjoint routes on its own, so a forest on the n vertices and the hyperedges together carries at most one edge fewer than it has nodes
- each hyperedge contributes r of those incidence edges, and that inequality rearranges directly into the bound

SCOPE Exact at $m = 2$ and at $m = 3$, where the same graph's cycle rank is bounded instead. From $m = 4$ it is open.



$$n = 9, r = 3, m = 2$$

EXTREMAL

Multihypergraph · edge

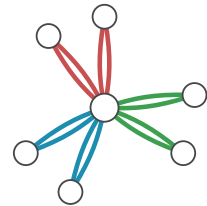
$$L_m^{(r)}(n)$$

UPPER 6 **PROVED** the same hyperedge cut, unchanged by repetition

LOWER 6 **PROVED** three pairs, each hyperedge taken $m - 1$ times

- the cut argument never asks whether hyperedges are distinct, so repetition leaves the bound exactly where it was and only changes which shapes can reach it
- at the smallest sizes it genuinely buys something: two copies of the only available triple reach a count no simple hypergraph can

SCOPE Exact whenever the pairs divide the leaves evenly, which is the case here.



$$n = 7, r = 3, m = 3$$

EXTREMAL

Multihypergraph · vertex

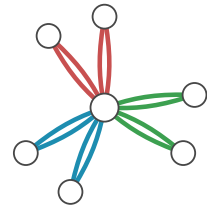
$$K_m^{(r)}(n)$$

UPPER 6 **PROVED** the incidence graph again, now by its cycle rank

LOWER 6 **PROVED** the same hypergraph

- at $m = 3$ the step is to bound the incidence graph's cycle rank by decomposing it at its cut vertices and separating pairs
- repetition buys nothing at $m = 2$, where two copies of one hyperedge are already two separated routes, but it does buy something from $m = 3$

SCOPE Exact at $m = 2$ and $m = 3$. Larger thresholds are open.



$$n = 7, r = 3, m = 3$$

EXTREMAL

Directed simple hypergraph · arc

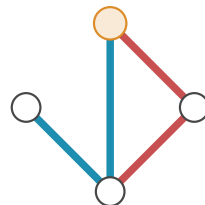
$$\ell_m^{\text{dir}(r)}(n)$$

UPPER 8 **PROVED** enumeration at this size

LOWER 8 **PROVED** two outgoing hyperedges on every vertex

- every route out of a vertex has to start with one of that vertex's own outgoing hyperedges, so deleting those cuts it off and the outgoing count alone caps every pair it reaches
- that gives feasibility at any size; it is the matching upper bound that is available here only by enumeration
- one tail is drawn in orange, and the other three repeat it around the square

SCOPE Only the growth rate is settled in general, by a shadow digraph that carries the hypergraph's route cap onto two-step routes.



$$n = 4, r = 3, m = 3$$

EXTREMAL

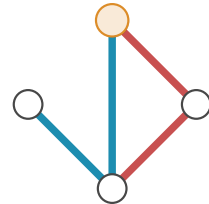
Directed simple hypergraph · vertex

$$k_m^{\text{dir}(r)}(n)$$

UPPER 8 **PROVED** enumeration at this size

LOWER 8 **PROVED** the same hypergraph

- the outgoing cut never looks at interior vertices, so it establishes the arc bound and the vertex bound in one go
- the shadow argument behind the general growth rate re-runs under the stricter separation for the same reason



$$n = 4, r = 3, m = 3$$

EXTREMAL

SCOPE Whether the two separations keep the same exact value at larger sizes is open.

Directed multihypergraph · arc

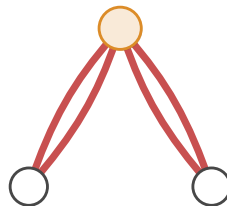
$$L_m^{\text{dir}(r)}(n)$$

UPPER 6 **PROVED** enumeration at this size

LOWER 6 **PROVED** $m - 1$ copies of the one hyperedge each tail has

- the outgoing cut caps each tail again, and at this size a tail has just one possible head set, the other two vertices, so repetition is the only way to grow at all
- lifting that limit on distinct head sets is the whole difference from the simple model

SCOPE Forward, backward and general orientations share the same growth rate. The exact values at larger sizes are open.



$$n = 3, r = 3, m = 3$$

EXTREMAL

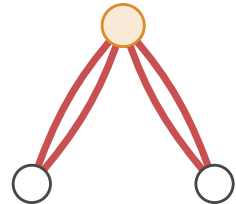
Directed multihypergraph · vertex

$$K_m^{\text{dir}(r)}(n)$$

UPPER 6 **PROVED** enumeration at this size

LOWER 6 **PROVED** the same hypergraph

- the outgoing cut gives both separations at once here too, so the arc witness needs no adjustment



$$n = 3, r = 3, m = 3$$

EXTREMAL

SCOPE Whether forward and general orientations agree once n passes r is open.